Thursday Warm Up: Comparators, Decoders and Encoders

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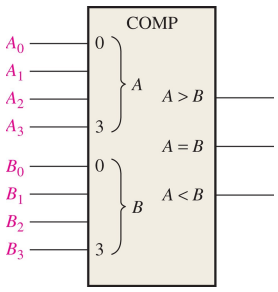


Figure 1: A comparator for two 4-bit numbers.

1 Functions of Combinational Logic: Comparators

1. (a) Draw a set of gates that yields the equality output of the comparator in Fig. 1. (b) Consider the $A > B$ output. Generate the truth table for the 2-bit case, in which we are comparing two 2-bit binary numbers. Develop the AND-OR logic for $A > B$ from the truth table. (c) Repeat, for the $A < B$ case.

2 Functions of Combinational Logic: Decoders and Encoders

1. Consider Fig. 2, in which the decoding function for 1011 is displayed. Generate all gate diagrams for the BCD codes. That is, for codes 0000, 0001, ... generate circuits that go HIGH to represent the digits 0-9.

2. Consider Fig. 3. Explain why a HIGH on the 9-wire will generate $A_3A_2A_1A_0 = 1001$. What happens if each of the other wires have a HIGH?

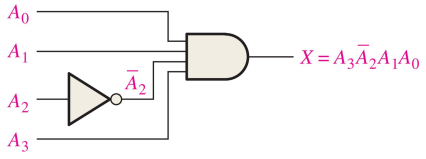


Figure 2: The decoding function for 1011.

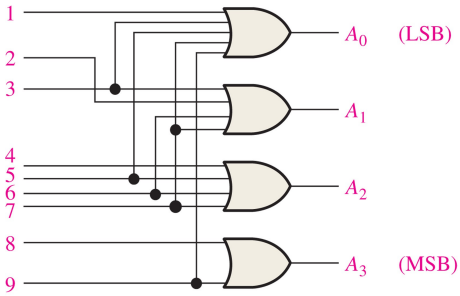


Figure 3: A decimal to BCD encoder.